

EXPERIMENT-05

AIM

To write a Python program using NumPy to calculate the average fuel efficiency of car models and determine the percentage improvement in fuel efficiency between two selected car models.

ALGORITHM

1. Start the program.
2. Import the NumPy library.
3. Create a NumPy array `fuel_efficiency` storing fuel efficiency (mpg) of various car models.
4. Calculate the average fuel efficiency using `np.mean()`.
5. Select two car models (e.g., model A and model B) by indexing the array.
6. Compute the percentage improvement using the formula:

$$\text{Percentage Improvement} = \frac{\text{ModelB} - \text{ModelA}}{\text{ModelA}} \times 100$$

7. Display the average fuel efficiency and the percentage improvement.
8. End the program.

PROGRAM

```
import numpy as np

# Fuel efficiency of different car models (in mpg)

fuel_efficiency = np.array([18, 22, 25, 30, 28])

average_eff = np.mean(fuel_efficiency)
```

```
# Compare model 1 and model 4 (indexes 0 and 3)

model_a = fuel_efficiency[0]

model_b = fuel_efficiency[3]

percentage_improvement = ((model_b - model_a) / model_a) * 100

print("Average fuel efficiency:", average_eff)

print("Percentage improvement from model A to model B:", percentage_improvement)
```

SAMPLE INPUT

[18, 22, 25, 30, 28]

SAMPLE OUTPUT

Average fuel efficiency: 24.6

Percentage improvement from model A to model B: 66.66666666666666

RESULT

The program successfully calculated the average fuel efficiency and determined the percentage improvement between two selected car models using NumPy array operations.